

Script of the Design Course

Design Course: RF- and Microwave Circuits and Systems

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Version 12.0



EEH, Fritz-Hüttinger chair energy-efficient
high-frequency electronics
Freiburg University

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Introduction

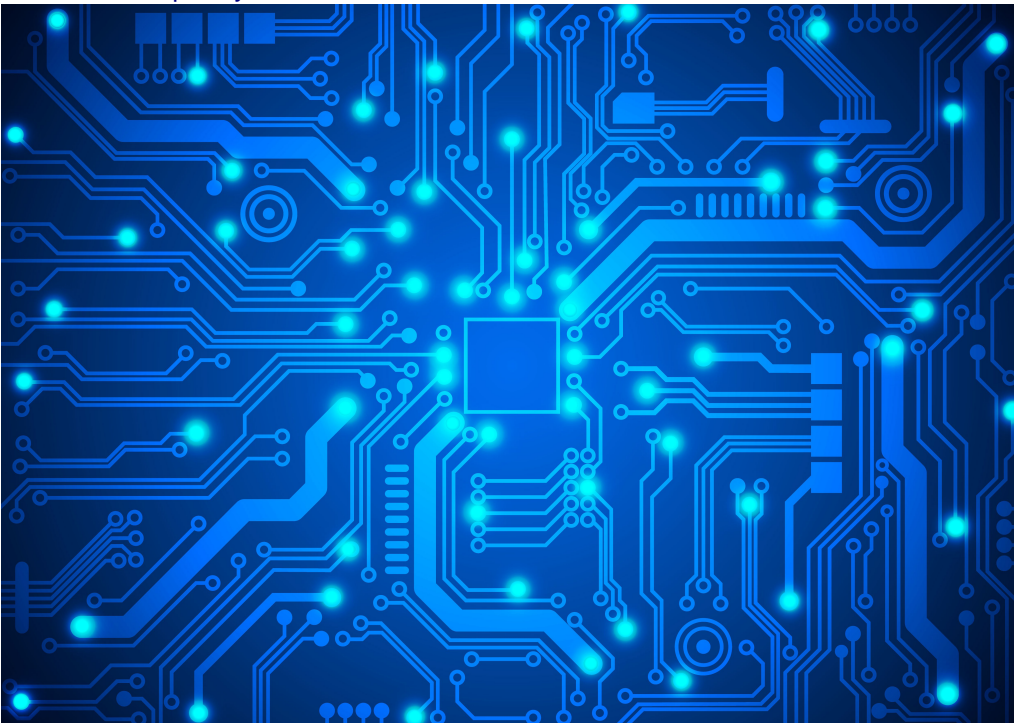


Introduction



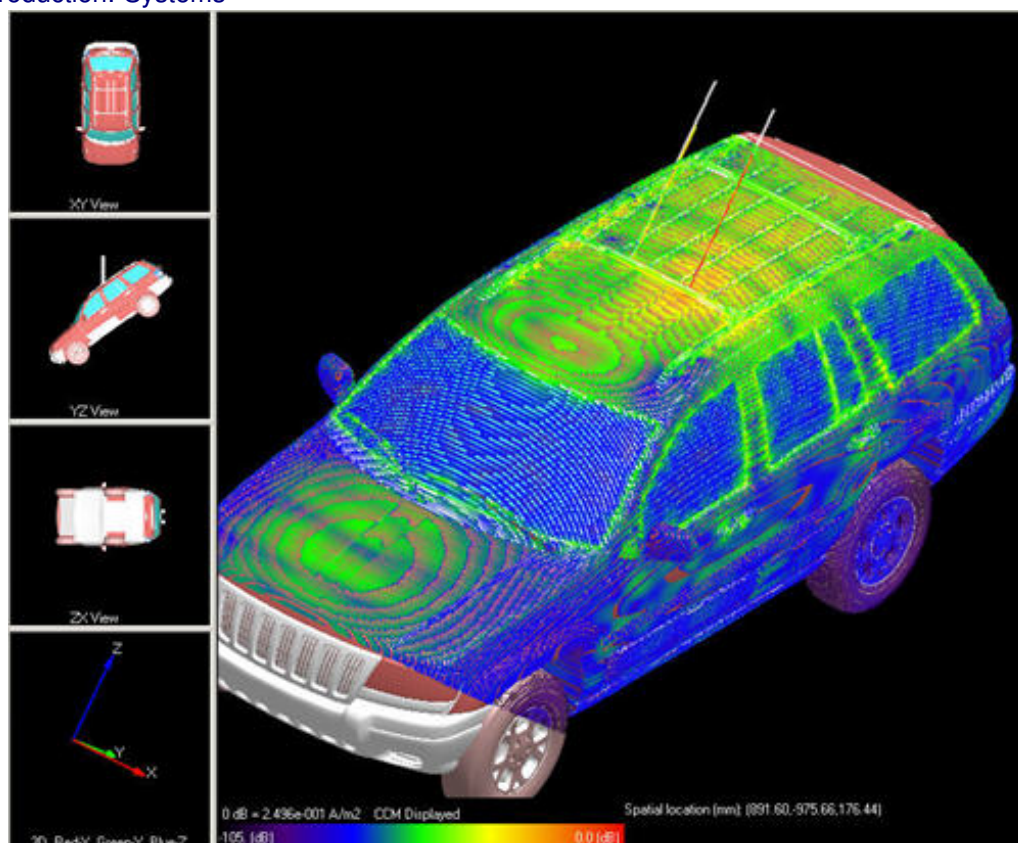
1.4

Introduction: Complexity



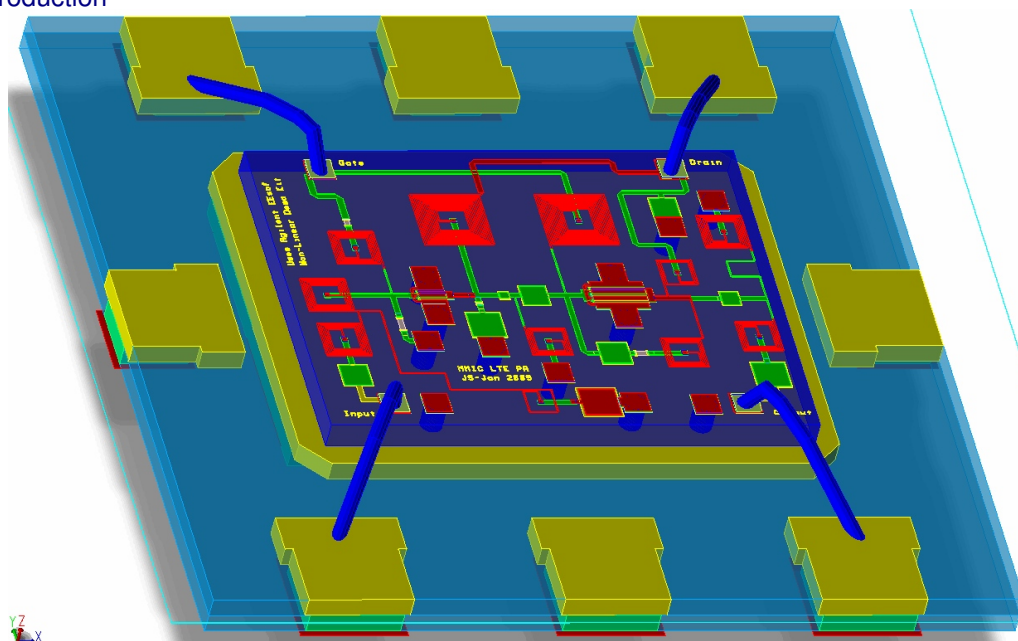
1.5

Introduction: Systems



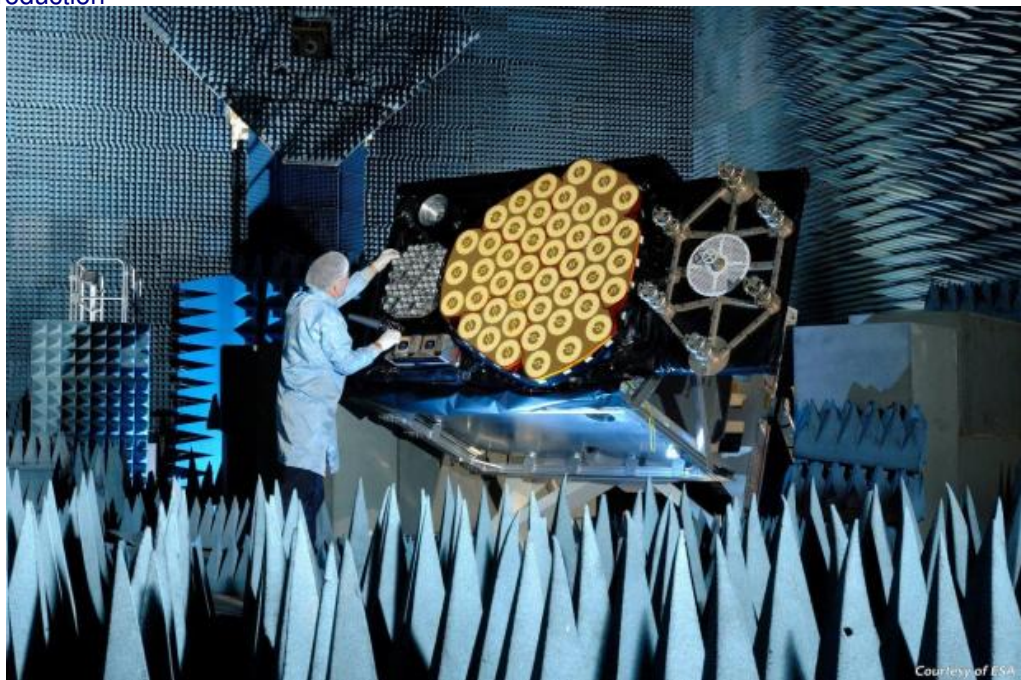
1.6

Introduction



!!!! Frequency domain !!!! vs. Cadence Virtuoso: Time domain

1.7



Dates with exercise

- Mo 1.) 20.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 2.) 27.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 3.) 4.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 4.) 11.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 5. 18.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 6. 25.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 7. 1.06.2026 no lecture, pentecost break SR 00 014 (G.-Köhler-Allee 078)

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Dates with exercise II

- Mo 8. 8.6.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 9. 15.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 10. 22.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 11. 29.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 6.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 13.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 13. 20.07.2026 Presentation of final results from competition 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)

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Date and Location

Dates with exercise

1. Mo, 20.4.2026: 16-17.00 h Introduction and group formation
2. Block 1: Mo., 27.4.2026: (issue block 1a on ILIAS)
3. Block 1: Mo., 4.5.2026: (issue block 1b on ILIAS)
4. Block 2: Mo., 11.5.2026 (issue block 2a on ILIAS), (Deadl. Block 1)
5. Block 2: Mo., 28.5.2026 (issue block 2b on ILIAS)
6. Block 3: Mo., 8.6.2026 (deadline block 2 on ILIAS), (issue block 3a on ILIAS)
7. Block 3: Mo., 15.6.2026 (issue block 3b on ILIAS)
8. Block 4: Mo., 22.6.2026 (Deadline block 3), (issue block 4a on ILIAS)
9. Block 4: Mo., 29.6.2026, (Issue Block 4b, COMPETITION !!!)
10. Block 5: Mo. 6.7.2026 (deadline block 4), (issue block 5a on ILIAS)
11. Block 5: Mo., 13.7.2026 :(issue block 5b ON ILIAS)
12. Mo., 20.7.2026, Final presentations based on 4b (deadline block 5 on ILIAS)

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Rules of the Exercises

- 5 blocks Exercises (every one/two weeks) with overall max. 10 tasks in a given format...
- First: starting 27.4.2026... (two weeks to do, sometimes more)
- 4 blocks to be completed successfully (with 50% correct) (three people per group)
- Voluntary Competition: Final presentation (5 mins) of the group to the plenary based on a task
- Exercises are given out on ILIAS and also submitted there.
- **Timing is day of the deadline, 16 h**

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1 Abstract

The students will be enabled to understand, design and layout modern RF- and microwave components and systems by means of the electronic design environment Advanced Design System including the two- and three dimensional electromagnetic simulators Momentum and EMPro 3D. The detailed use of a complex RF-software environment is a dedicated target of this course. This includes the numerical analysis of complex passive and active devices, the design and layout of hybrid and integrated circuits, and their packaging and signal flow. The students will be competent to design and layout passive and active RF-structures including packages and interconnects and circuits of relevance to everyday communication and sensing. The competence includes in-depth understanding and treatment of complex microwave systems and of general system design including the treatment of complex modulated signal flows.

Introduction

Materials and Script

The material is provided at the beginning and during the lecture.

Email: ruediger.quay@inatech.uni-freiburg.de

- Foils: Intro materials from Keysight
- ADS Video Library on www.youtube.com
- Script (supporting materials)
- Exercises

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Videos

1. Watch ADS videos on youtube using ADS : www.youtube.com , Keyword: ADS Video Library
2. Basics and setup:
3. <https://www.youtube.com/watch?v=04BfajzXnmM&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz>
4. Understanding Workspaces:
5. <https://www.youtube.com/watch?v=7jNm400BaiE&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=2>
6. The basics of schematics:
7. <https://www.youtube.com/watch?v=wqyyOCzxGS0&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=4>
8. Data displays:
9. <https://www.youtube.com/watch?v=jIg10RihLQk&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=6>

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WWW Support 1

ILIAS Support

<https://ilias.uni-freiburg.de>

search for: Design Course: RF- and Microwave Systems

password: design2026

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WWW Support 2

This exercise is recorded, attendance is, however, required



1.16

Windows Access to the Uni Network

- Get login at university, get additional TF login
- Install VPN from University server
- use ssh from linux
- If you start from Windows:
- Use, e.g., xming as windows X-server (for export of the display)
- Install Putty ssh client on windows

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Form groups of Three

- The exercises are submitted from a group of three people
- Find two partners !!!
- Send your group to ruediger.quay@inatech.uni-freiburg.de, including TF login name.
- Exercises start april 27th 2026 !

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Install an ADS Version on a local computer

Last Updated: 05 March 2020

ADS - Windows® Operating Systems	
Product	Windows
ADS 2020 / ADS 2020 Update 1.0 / ADS 2020 Update 2.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2019/ ADS 2019 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2017/ ADS 2017 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2016.01 64-bit Operating Systems	Windows 10 (64-bit)* Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2015.01 64-bit Operating Systems	Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)

You can work on the aragorn but local has proven to be easier...

Organize your data in a good structure

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Login to Application Server for the license

From Remote: Create RAS Access in Mylogin

Go into University net with VPN: check VPN Uni Freiburg

In case you use Windows to do that: use putty and install xming as windows X-server

Application Server: `aragorn.imtek.uni-freiburg.de`, `ssh-X-1 (yourlogin) aragorn.imtek.uni-freiburg.de`

Shells usable: `bash` and `tcsh`

Start: `ads` after using `ssh` to the server

Licence Server will be asked: `27010(at)frodo.imtek.uni-freiburg.de`

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Backup Server !

Application Server: `samweis.imtek.uni-freiburg.de` (only via VPN)

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Recommendation

Use Licence server only, install ADS on your local device ! Use ADS 2023 Update 1.0

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The following references may be useful to provide more detailed information.

References

- List of useful ADS Videos
- Keysight Software Support: <http://www.keysight.com/>
- Introduction to ADS, ADS_Basics.pdf (ILIAS)
- Momentum Momentum_Basics.pdf (ILIAS)

1.23

2 Passive IC Components: Block 1 Part B

2.1 Tasks 4: MMIC Passive Structures (Block 1 Part B)

2.2 Educational Targets

Educational Targets: MMIC Passive Structures

- Understand passive integrated components in the project...
- Simulate more complex structures other than a transmission line...
- Organize the single component description and compare to the complete description...
- Understand the creation of real matching network on layout level..
- Understand the need to be precise...

3.25

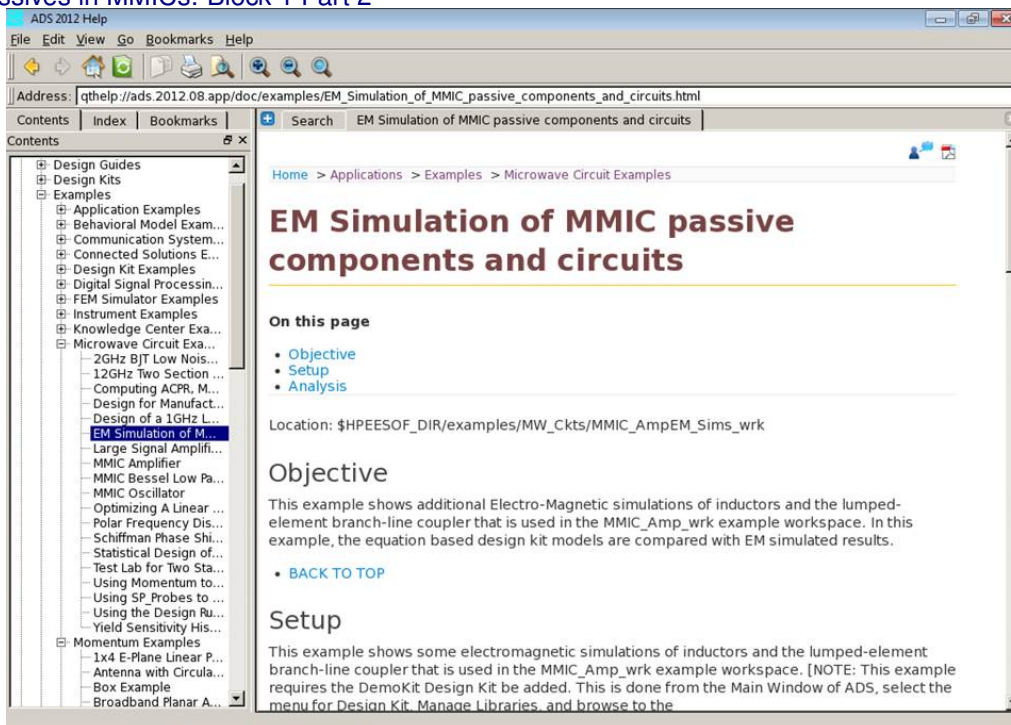
2.3 Real Lines in MMICs: Block 1 Task 4

Momentum Examples: Build a Structure

1. https://www.youtube.com/watch?v=Y4H0QsXN-aE&list=PLtq84kH8xZ9GM_bQbYpqrdsQ0tc-ZfFc&index=2
2. https://www.youtube.com/watch?v=0ITHN5pEXYQ&list=PLtq84kH8xZ9GM_bQbYpqrdsQ0tc-ZfFc&index=3
3. <https://www.youtube.com/watch?v=4lybtDEc7fw>

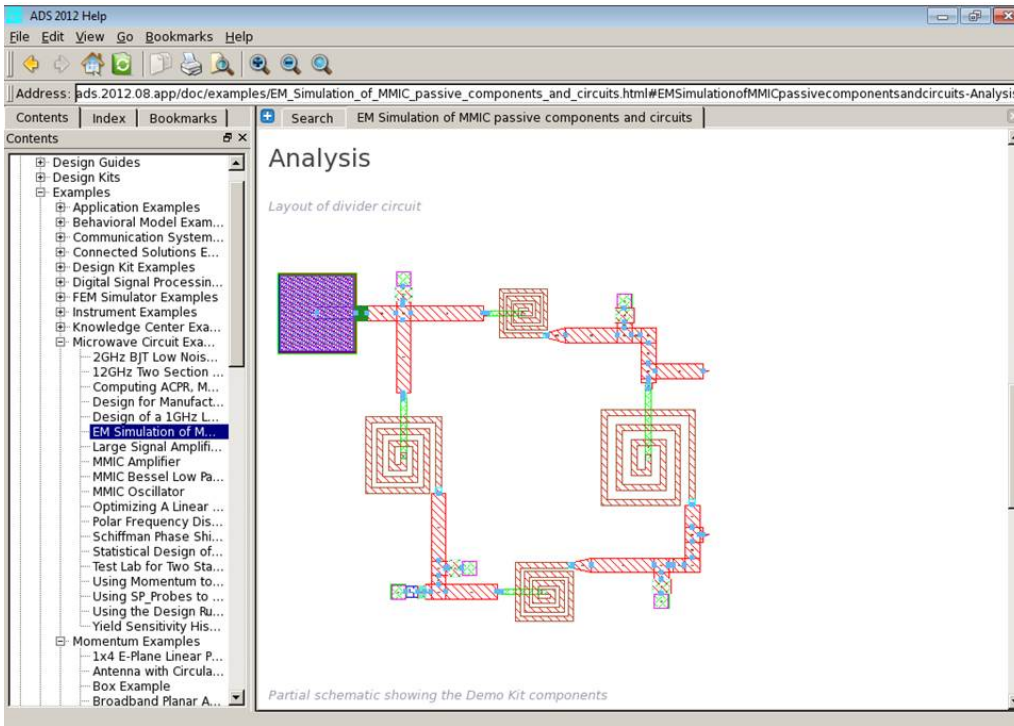
3.26

Passives in MMICs: Block 1 Part 2



3.27

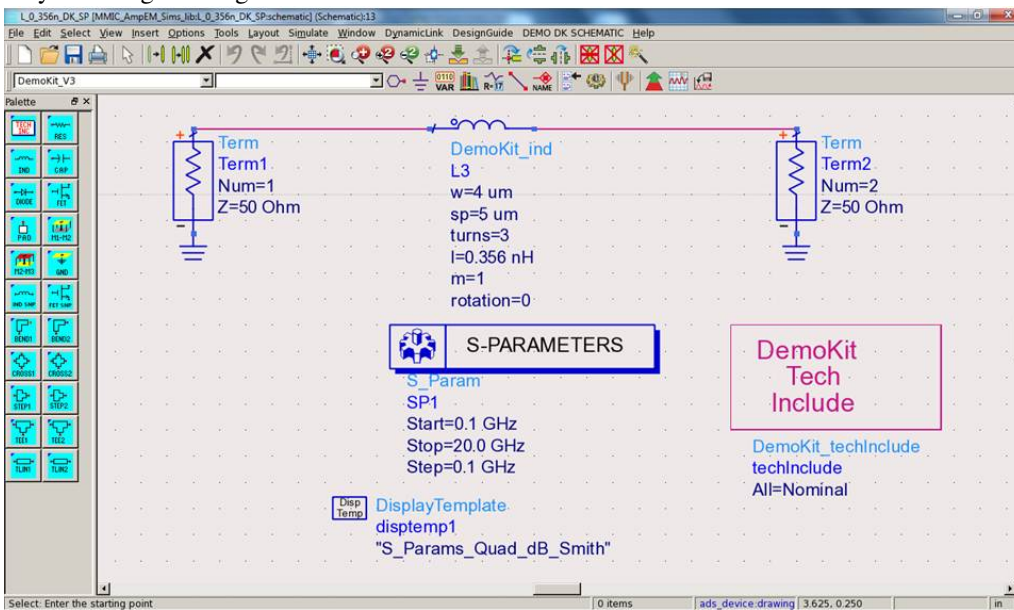
Schematic of a MMIC



3.28

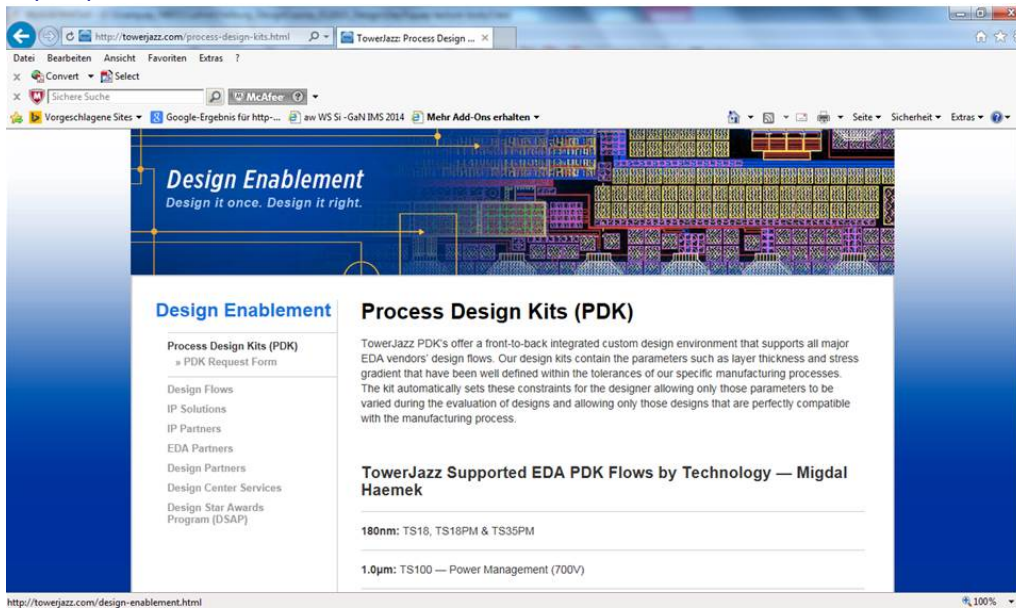
Integrated Components from Design Kits (Process Design Kit: pdk)

Other components will be included given by the semiconductor manufacturer. The fab will talk to you through a design kit !



3.29

Example: pdk



3.30

2.4 Tasks 4: EM in MMICs (Block 1 Part 2)

Tasks 4: MMICs: Monolithically Microwave Integrated Circuits

1. Unzip the project in MWckts/MMICAMPEM_Sims_wrk (You need to include the EM Files)
2. Go to readme !
3. Perform an S-parameter simulation through Momentum (MOM) of two inductors of your choice from the pdk for $f=0\ldots3$ GHz. How do you expect a pure inductor to behave (One plot, three sentences, 5 points).
4. Compare the results to the results of the design kit models used (One plot, three sentences 5 points).
5. In three sentences: what could be reasons, why they are different ? (3 sentences, 5 points)
6. Create the first letter of your name by transmission lines and run a two port simulation from $0\ldots10$ GHz. Compare the results of the four S-parameters in one plot of the schematic simulation to a momentum simulation of the same. Explain any major differences you observe in three sentences (Layout, plot, three sentences, 10 points).

3.31

Format and points Task 4 (Block 1 part 2) Momentum (Sum of points: 25 points)

1. Number 3.)(One plot, three sentences, 5 points)
2. Number 4.)(One plot, three sentences 5 points).
3. Number 5.)(3 sentences, 5 points)
4. Number 6.)(plot layout, 5 points).
5. Number 7.)(Layout, plot and 3 sentences) (10 points)

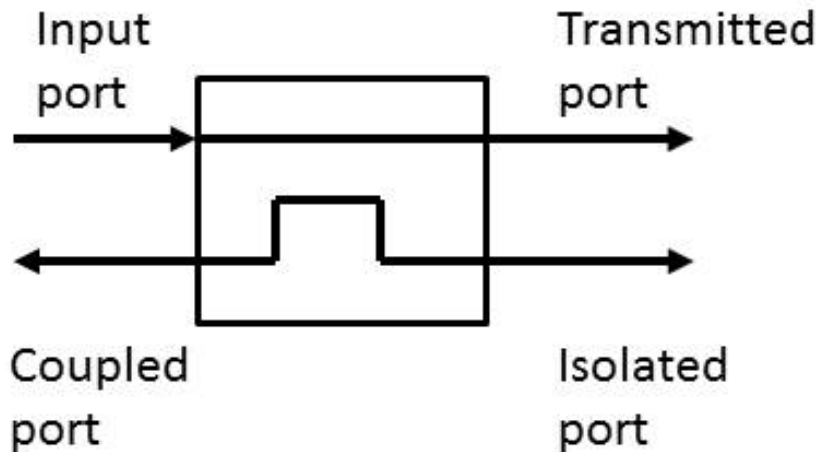
3.32

3.33

4. may 2026

2.5 Task 5: Designing a Coupler: 2.5 D and 3D Simulation (Block 1 Part 2)

Coupler: Principle



Design a coupler with three ports from two MSL lines...

3.34

Directional Coupler: Design Intro

ADS 2012 Help

File Edit View Go Bookmarks Help

Address: qthelp://ads.2012.08.app/doc/examples/Directional_Coupler.html

Contents Index Bookmarks

Search Directional Coupler

Setup

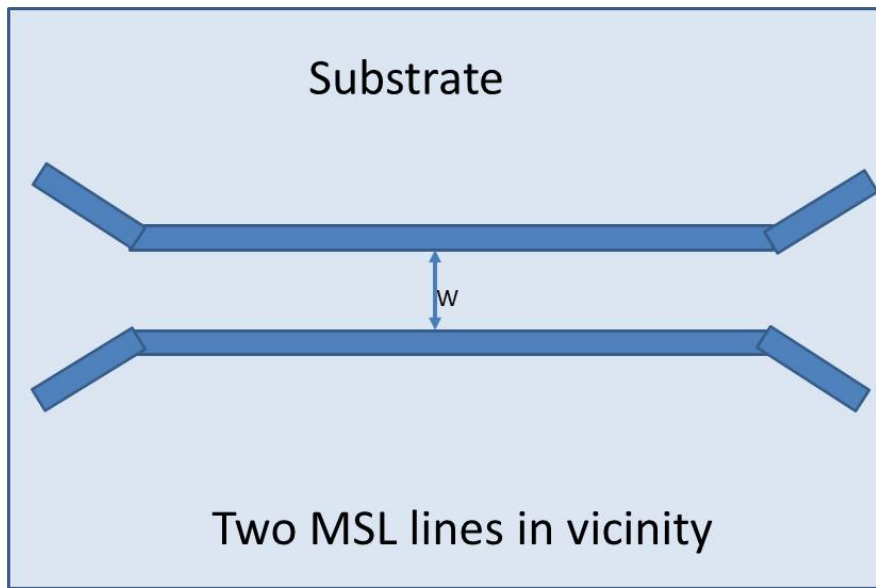
- **ideal_cplr**: The design begins with selection of even Z_e and odd Z_o impedances for -20 dB and coupling with 50 Ohm input. The reference paper is mentioned in the design. The values of impedances obtained from tables is implemented with Ideal coupled line CLIN and schematic simulation is carried out to check whether the choice of Z_e and Z_o is giving the required performance.
- **cplr_strp**: The Line Calculator is used to find the width of the line W and gap W_o for Offset Stripline SOCLIN based on calculated Z_e and Z_o in previous step. The schematic circuit is developed and circuit simulation is carried out to check required coupling of -20dB.
- **cplr_strp_1**: From Momentum simulation, it is found that coupling is -19.78 db instead of the required -20dB. Therefore, the circuit is actually over coupling than the required level. The optimization for gap W_o (which is responsible for coupling) is performed at the circuit level. After the optimization, the value of W_o is found to be more than used in cplr_strp. The momentum simulation shows coupling of -20 dB now.
- **cplr_strp_1_input**: The input section is added to the circuit to make it compatible with 50 Ohm input connector. Again circuit simulation, Momentum Simulation and Optimization is carried out at this level.
- **cplr_strp_1_input_spacers**: The PTFE Spacers are added to the circuit. FEM simulation is carried out. The circuit cplr_strp_1_input is used and PTFE spacers are added. The coupling is found to be increased because of the presence of PTFE Spacers(see cplr_strp_1_input_spacers_emds.dds). The optimization is again carried out for W_o to bring back the coupling at -20 dB level.
- **Comparison_emds_mom.dds**: Comparison of FEM and Momentum result when the gap W_o is not optimized in the presence of PTFE spacer.
- **Comparison_emds_opt_mom.dds**: Comparison of FEM and Momentum result when gap W_o is optimized in the presence of PTFE.

Figure 1: Directional Coupler and Substrate Stackup

3.35

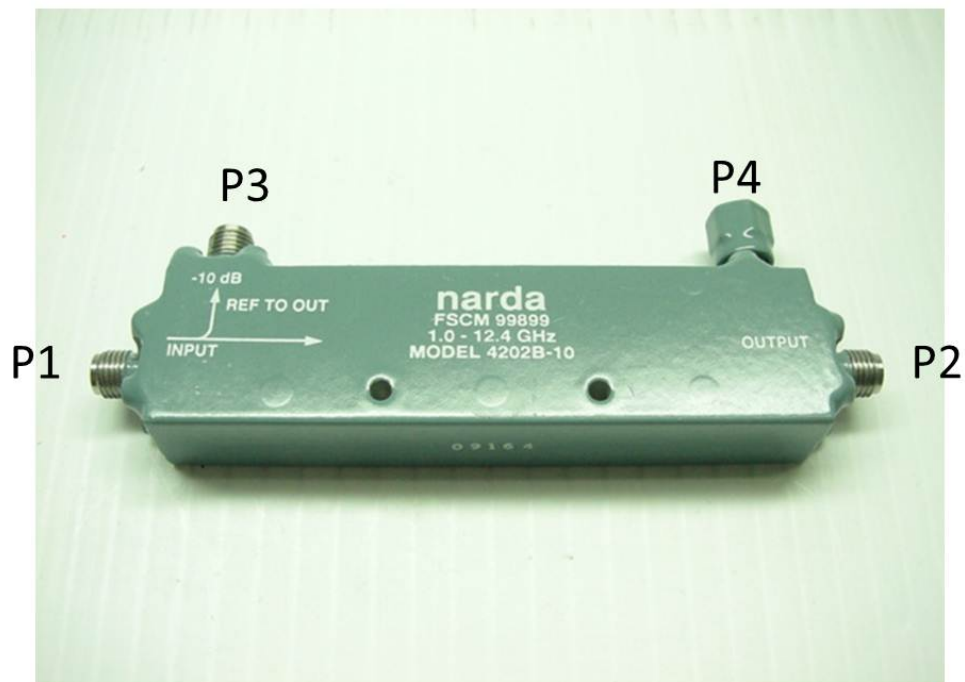
Coupled Lines

Coupled Lines: Coupler



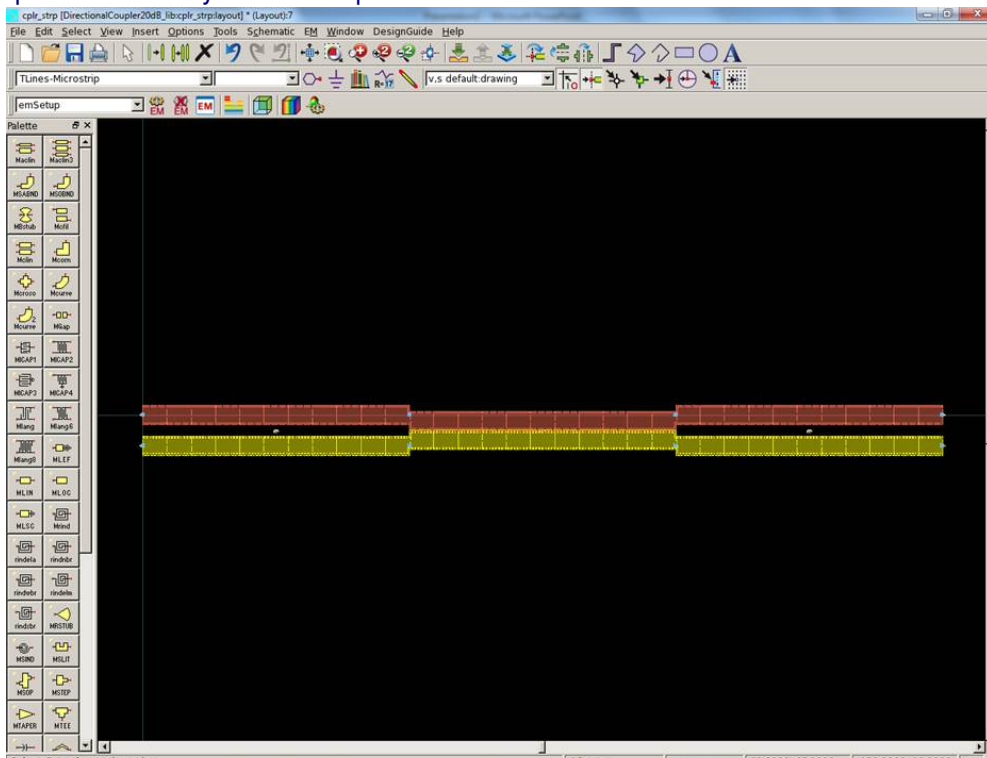
3.36

Coupler Example



3.37

Coupler: MOM Analysis of a Coupler



3.38

2.6 Tasks 5: Coupled Lines Coupler (Block 1 Part 2)

Tasks Coupler (Block 1 Part 2)

1. Goto readme of the project examples...FEM...directionalcoupler20dB in examples
2. Explain in four sentences, what are the main ports (name all four) and quantities (name three) of a coupler device (5 points).
3. Go to `clr_strp_1`. Perform a MOM analysis with Momentum of the coupler, plot the results for a.) default and b.) for a reduced WO (adjacent distance between the lines only in the middle section of the structure !) changed in such a way that the maximum coupling increases S31 from -20 dB (default) to -6 dB. (2 plots, 5 points).
4. Compare the two results for the two relevant S-parameters S11 and S31 between 0..3.7 GHz. HINT: You can reduce the number of frequency steps in the simulation to save time in the MOM (1 Plot, 5 points).
5. Go to one of the FEM simulations of this example. Save the results file (do not overwrite it, give a new name !!!). Run the FEM simulation for the 20 dB default for the frequency of 1.3 GHz at least three levels. You can stop after that. Plot the Sij of the coupler (1 plot, 5 points).

3.39

Format and points Task 5 (Block 1 part 2) Momentum (Sum of points: 25 points)

1. Number 2.) Name the main ports (name all four) and quantities (name three) (5 points).
2. Number 3.) (2 plots, 5 points).
3. Number 4.) (1 Plot, 5 points)
4. Number 5.) (1 plot, 5 points).

3.40